THE ANATOMY OF A STORY

Make the judge understand, believe and remember.

Naiier Eldeeb

NASA Space Apps Cairo · 30 October 2026

01 **UNDERSTAND**
A clear idea

02 **BELIEVE**
Evidence inside the story

03 **REMEMBER**
One message that stays


CNET
One more thi

NASA SPACE APPS CAIRO / STORYTELLING WORKSHOP



WORKSHOP SLIDE 01

ONE WEEK. ONE SHARED PROJECT.

A college project. A shared folder. Another search every time.

7

DAYS

Working together at college

41

FOLDERS

200+

FILES

MORE THAN 200

Every time we needed a file, we had to search again.

Source: Naiier's recollection of the Cells college project.

We made them.

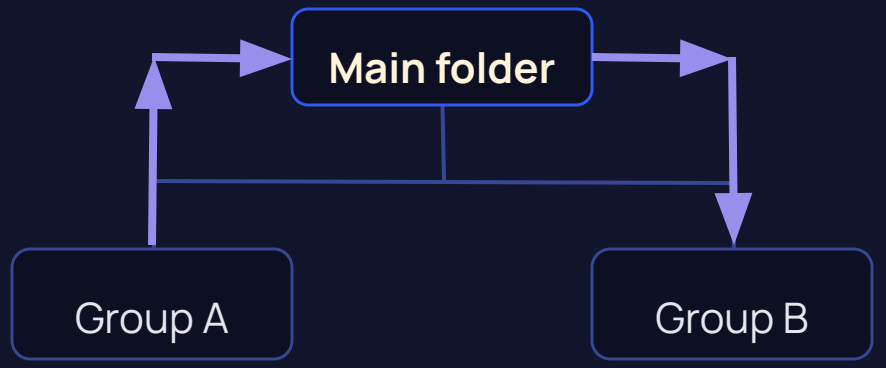
We still had to search for them.

Has this happened to your team?
Raise a hand.

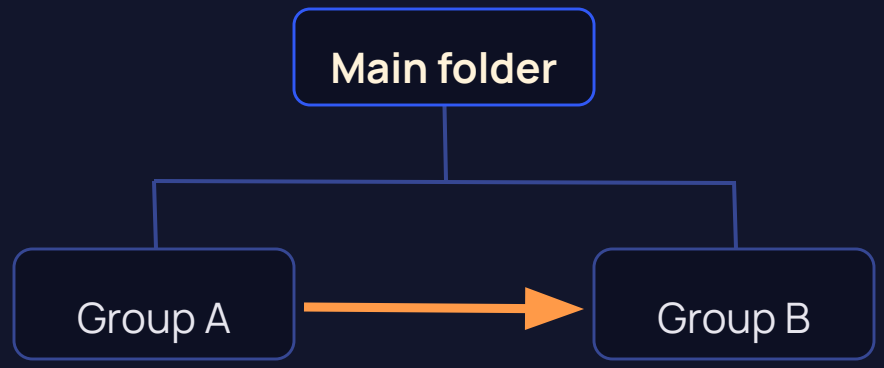
CHANGE THE ROUTE BETWEEN GROUPS

Cells · a UI/UX concept inspired by relationships in living systems.

A HIERARCHICAL ROUTE



THE CELLS PROPOSAL



Main cell
The main folder.

Connected groups
More than one route between groups.

A design proposal
A concept, not a tested speed result.

Educational reconstruction of the concept; not the original interface.

Show Why One Step Leads On

A sequence becomes clearer when its connections are visible.

01	**Chronology**	College project → many files → a new interface.
02	**Cause and consequence**	Repeated searching frustrated the group → navigation became a design question.
03	**Check the connection**	Can you explain why this next step follows?

Four Things to Leave With

Draft them, say them aloud, and test what someone else understands.

One sentence

The project and its intended purpose.

Six-beat storyboard

A sequence of messages, visuals, and evidence.

60–90-second opening

A rehearsal target for this workshop.

Three takeaways

What you want the listener to remember.

THE TWO-HOUR JOURNEY

Understand the choices. Watch a rewrite. Build and test your own.



0 min 120 min

00–12
The Cells story

12–24
How explanations work

24–39
Storyteller Examples

39–51
Space Apps cases

51–71
Live rewrite

71–101
Your practice

101–113
Feedback + revision

113–120
Checks + questions

Working run of show · bar widths show duration · 30 minutes reserved for practice.

ONE PROJECT. THREE STARTING POINTS.

The facts stay the same. The listener changes what needs explaining first.

01

EVERYDAY USER

“Can I get to the file I need for this task?”

Show the task and the route.

02

UX DESIGNER

“How are the groups connected?”

Show relationships and interaction.

03

PROJECT EVALUATOR

“What did you build, and what supports it?”

Show the problem, prototype and proof.

Which question would your opening answer?

Write the First Sentence

Explain Cells to a teammate who missed the project.

Choose what they need

What must they understand before the interface makes sense?

01

Write one sentence

Use the known experience: group work, cluttered files, repeated searching.

02

Test the sentence

Ask a partner to name the problem they heard.

03

Try ABT If It Helps

An optional drafting tool: And, But, Therefore.

01

And: the situation

We worked as a group and created many files.

02

But: the difficulty

Finding the files we needed became frustrating.

03

Therefore: the response

We began rethinking how to move between file groups.

OPTIONAL TOOL / Read the finished sentence naturally.

Match the Claim to the Stage

Help the audience see what you know and what remains open.

STAGE 01

Proposal

We propose connected groups as a different navigation approach.

STAGE 02

Prototype

This model demonstrates a particular route or interaction.

STAGE 03

Tested result

Describe the test, participants, conditions, and observed outcome only when available.

The wording must fit the evidence available at that stage.

Al Daheeh: Begin with a Problem

How Do You Write an Al Daheeh Episode? | 00:03–00:33



A familiar situation
An applicant wants a job.

An unexpected obstacle
The company hires only AI.

Your question
What do you want to know next, and what matters to this person?

Official episode: Al Daheeh, Museum of the Future

Let the Concept Answer a Question

A later excerpt from the same episode | 05:13–05:43



The question develops
Can a computer learn to write?

The concept enters
Machine learning becomes part of the explanation.

Apply the choice
Beside your hardest term, write the question it helps your audience answer.

Official episode: transition into machine learning

Fi El Hadara: Start with Familiar Behaviour

The Story of AI: Can Machines Think? | 01:36–02:09



Everyday behaviour
Naming a chatbot, asking about work, and confiding in it.

A comparison opens
The discussion moves toward comparing it with a human friend.

Pause and identify
Which familiar detail leads you toward a larger question?

Official episode: Fi El Hadara, AlarabyTube

Study the Choice, Then Adapt It

Turn your observations into decisions for your own explanation.

01	**Choose an entry point**	Use a situation your audience can understand.
02	**Create a useful question**	Let the opening establish something the explanation can address.
03	**Keep your own voice**	Adapt the reasoning to your facts, audience, and time; test what listeners understand.

Al Daheeh example discussed in the lab · Fi El Hadara example discussed in the lab

What makes a story ?

Cells: a new workshop wording of the same experience.

Character / Goal / Conflict / Stakes / Action / Result and conclusion

An Opening Creates a Next Step



An Opening Creates a Next Step



Three projects. Three storytelling choices.

Study one decision you can test in your own explanation.

01

A Solar Tale

How does choosing children change the explanation?

02

TerraTales

What can a family analogy clarify—and where does it stop?

03

Henry's Farm Adventures

How can a decision give the next piece of information a purpose?

A SOLAR TALE: THE STORY CARRIES SCIENCE

HerCode Space · 2025 Global Winner · Best Use of Storytelling

01 NARRATOR

The Sun tells the story in first person.

02 STRUCTURE

The story is organised in chapters.

03 EVIDENCE

NASA and NOAA media appear inside the story.

04 YOUR DESIGN TEST

What must stay accurate when the explanation gets simpler?

NASA: A Solar Tale, 2025

TerraTales: give information a shared context

Cairo · 2024 Best Storytelling



GENERATED TEACHING ILLUSTRATION
Not original project artwork.

Story setting

Mother Earth meets regional characters; the project uses Egypt, Brazil, and Germany.

Discuss with a partner

What does a family analogy make easier to follow?

Name the boundary

Which scientific statements still need their own sources?

Story → context → evidence

Generated teaching illustration; not project artwork.

Henry’s Farm Adventures: make sequence matter

Zumorroda-X · 2025 Art & Technology



GENERATED TEACHING ILLUSTRATION
Not original project artwork.

Situation

The player follows a farmer facing environmental challenges.

Decision

Mini-games connect the journey to choices and learning.

Next information

What must the player understand before the next decision?

Situation → decision → result

Generated teaching illustration; not project artwork.

Keep the decision you can explain

Three ways to help someone follow an idea.

01	**Choose the audience**	A Solar Tale: decide what needs a picture, an example, or a definition.
02	**Connect information**	TerraTales: use an analogy and state its limits.
03	**Order the experience**	Henry's Farm: prepare the learner for a decision.
04	**Apply**	Which choice would help your explanation first?

What does a feature list leave unanswered?

Our teaching draft—not the team's original pitch.

A website with stories, charts, a game, climate data, and interactive sections.

Fact card
Climate story: Egypt, Brazil, Germany. Choice-based game.

Source stated by the team
U.S. Greenhouse Gas Center.

Evidence limit
No measured learning or behaviour result in this card.

Choose who needs this explanation

Our teaching choice—not the team's original audience claim.

Audience

A first-time visitor without a climate background.

Purpose

Help them connect a climate fact with its place and source.

Test the choice

What do they already know? What do they need next?

TEACHING CHOICE / Change the audience; reconsider the opening.

TerraTales: exercise fact source

Give the features a clear purpose

An educational rewrite using the same fact card.

Before

A website with climate stories, charts, a game, and interactive sections.

After

TerraTales introduces climate information through a story and interactive choices, aiming to help first-time visitors explore what it means.

Keep the distinction

An intended benefit still needs testing.

TerraTales: exercise fact source

SIX BEATS. ONE JOB PER BEAT.

A storyboard makes the sequence visible before you polish the slides.

01 HOOK + PROBLEM

What needs explaining?

02 USER + STAKES

Who needs to understand — and why?

03 DESIGN INSIGHT

What connects the information?

04 SOLUTION + DEMO

What will the visitor experience?

05 EVIDENCE + LIMITS

What supports the explanation?

06 IMPACT + NEXT TEST

What should change? How would we check?

TerraTales: exercise fact source

Match each claim to its evidence

Keep the source beside the information it supports.

What we can describe

The project includes a choice-based game.

SOURCE: TEAM DESCRIPTION

What needs a test

“The website changes behaviour.” Define the behaviour and measure it.

REQUIRED: A DEFINED BEHAVIOUR TEST

For every chart Identify the source, measure, units, and period. Missing data is not permission to invent a chart.

Rehearse the opening, then listen

Educational rewrite · 60–90 seconds is a rehearsal target.

A climate number can leave a visitor asking: what does it mean for this place? Let's follow one part of TerraTales, see what it shows, and find the data behind it.

Listener check
What question are you waiting to have answered? What evidence do you still need?

KEEP THE FACTS. CHANGE THE ENTRY POINT.

Cells · an educational rewrite of the concept, using the same underlying facts.

BEFORE / FEATURES FIRST

An educational climate website featuring a Mother Earth story, climate data from Egypt, Brazil, and Germany, and an interactive choice-based game.

AFTER / A QUESTION TO FOLLOW

What do climate changes actually mean for different places, and how can storytelling help us understand the data behind them?

A clearer explanation is a communication choice.

It does not prove that the interface is faster or easier to use.

Reveal what the explanation needs next

Every movement should explain a relationship or a change.

01 QUESTION	02 CONNECTION	03 CHANGE
Start → ? → Target	Start → Route → Target	What changed?

Sequence

Show the question, then reveal the connection that answers it.

Live editing

Let the audience see what changed and why.

Comparison

Keep the same facts and reference points visible.

Shareable version

Use a readable final state with sources and enough context to stand alone.

Penn State: slide-design research summary

Your 30-Minute Story Lab

Use the supplied past-season case. Keep every claim within the fact card.

30

MINUTES

Work together

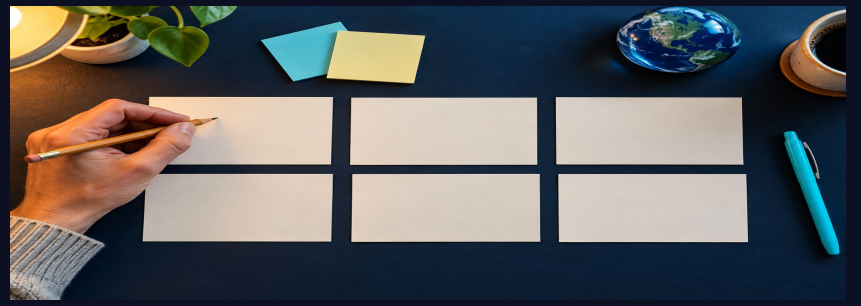
A small group or a partner is enough. Paper works.

Make four outputs

One sentence, six beats, one opening, three intended takeaways.

Keep your work

Every participant keeps a copy or photo of the group's outputs.



FRAME · 7m	STORYBOARD · 9m	REHEARSE · 6m	LISTEN · 5m	REVISE · 3m

Facts, Audience, One Sentence

First 7 minutes, including the brief setup.

7

MINUTES

Minutes 0–3

Read the fact card. Choose who will hear the explanation and what they already know.

Minutes 3–7

Write one sentence: who needs what, and how does the project aim to help?

Quick check

Read it to a partner. Ask them to describe the intended purpose.

FRAME · 7m | STORYBOARD · 9m | REHEARSE · 6m | LISTEN · 5m | REVISE · 3m

Build the Six Beats

9 minutes. For each beat: message, visual, evidence or limit.

1. Hook + problem

What needs explaining?

2. User + stakes

Who needs it, and why?

3. Design insight

What connects the idea?

4. Solution + demo

What can the viewer inspect?

5. Evidence + limits

What supports the claim?

6. Intended impact + next test

What would you check next?

FRAME · 7m | STORYBOARD · 9m | REHEARSE · 6m | LISTEN · 5m | REVISE · 3m

Test What Reached the Room

12 minutes: two 4.5-minute rounds, then a 3-minute debrief.

SPEAK	WRITE	FEEDBACK	REVISE

90 seconds

A volunteer group delivers its opening.

30 seconds

Listeners write purpose, recall, and missing information.

60 seconds

Hear specific feedback before the group responds.

90 seconds

Choose one change and try the revised part.

ROUND 1 · 4½ MIN → ROUND 2 · 4½ MIN → DEBRIEF · 3 MIN

THANK YOU !

NASA: 2025 Global Winners · NASA: 2024 Global Winners · TerraTales: original project submission